

Baohua Yang

Principal Architect | Enterprise AI, Agentic Systems & Distributed Platforms
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PROFILE

- Principal Architect and technical leader with 15+ years of experience taking enterprise AI, agentic systems, and distributed platforms from research to enterprise products.
- Designs and delivers enterprise AI and agentic systems, combining product strategy, system architecture, and hands-on engineering.
- Key contributor and maintainer across Hyperledger, OpenStack, and Docker; active mentor in the open-source community.

TECHNICAL SKILLS

- AI: multi-agent orchestration (LangChain/LangGraph), RAG, MCP, context management, evaluation & telemetry, data protection, LLM inference, fine-tuning.
- Languages & Systems: Python, Go, Kubernetes, containerized & distributed infrastructure.

EXPERIENCE

Principal Architect, *Oracle*

Aug 2017 – Present

- Led and built enterprise AI and distributed-system products for Fortune 500 customers, spanning a ~\$1B revenue portfolio and a 50+ engineer organization.
- Designed and implemented an enterprise AI-agent platform, improving accuracy from 30% to 90%, and cutting latency in half.
- Set technical direction across AI, security, data, and platform architecture; partnered with product and go-to-market teams to bring research into production.

Research Scientist, *IBM*

Jan 2013 – Aug 2017

- Pioneered the architecture and development of enterprise blockchain platforms and distributed-systems solutions for production use cases.
- Built production enterprise systems across blockchain, big data, and cloud, while driving the applied research that fed them.

OPEN SOURCE & COMMUNITY

Hyperledger Foundation TSC Member (now LF Decentralized Trust)

2017-2019; 2020-2021

- Steered technical strategy and roadmap development across all Hyperledger projects.
- Served as a core maintainer and developer, establishing foundational frameworks and architectural standards for decentralized technologies.

- Mentored open-source developers and new contributors.

EDUCATION

Ph.D. in Control Science and Engineering, Tsinghua University Sep 2007 – Jan 2013

Research focus: high-performance distributed systems and algorithms.

Built the first 10 Gbps unified threat management (UTM) system.

1st-class scholarship (2009) and IBM Ph.D. Fellowship Award (9 winners, 2011).

Visiting Scholar, EECS, University of California, Berkeley Sep 2010 – Sep 2011

Advanced distributed systems, including software-defined and data-center networking.

Bachelor of Science and Engineering, Tsinghua University Aug 2003 – Jul 2007

1st-class scholarship, 2006.

SELECTED PATENTS

- Methods and System to Support Long Context Management in AI Agentic System, 2026
- Enhanced NL2SQL AI Query System and Method, 2025
- Methods and System to Protect the AI Knowledge Base System, 2024
- [Methods and System of ID Generation for Transaction-Based System](#), 2020
- [Methods and System of Tracking Transactions for Distributed Ledger](#), 2019

SELECTED PUBLICATIONS

- *LazyCtrl: A Scalable Hybrid Network Control Plane Design for Cloud Data Centers*, Kai Zheng, Lin Wang, Baohua Yang, Yi Sun, and Steve Uhlig, IEEE Trans. Parallel Distrib. Syst., vol. 28, no. 1, pp. 115-127, 2017.
- *Keep Forwarding: Towards K-link Failure Resilient Routing*. Baohua Yang, Junda Liu, Scott Shenker, Jun Li and Kai Zheng, 33rd IEEE International Conference on Computer Communications (INFOCOM 2014), Toronto, Canada, April 2014.
- *Practical Multi-tuple Packet Classification using Dynamic Discrete Bit Selection*. Baohua Yang, Jeffrey Fong, Weirong Jiang, Yibo Xue and Jun Li, IEEE Transactions on Computers, pp. 424-434, Vol. 63, No. 2, Feb 2014.
- *Data-Driven Network Connectivity*. Junda Liu, Baohua Yang, Scott Shenker and Michael Schapira, 10th ACM Workshop on Hot Topics in Networks (HotNets-X), Cambridge, MA, November 2011.

SELECTED BOOKS AND TECHNICAL GUIDES

- Harness Engineering Guide and Agentic AI Guide (2026)
- Docker Container Technology in Action, China Machine Press, 2015 (4 editions).
- Blockchain: Principle, Design and Applications, China Machine Press, 2017 (2 editions, best award).